# Notes on Schur Decomposition

Handwritten Notes Conversion

## Introduction

When I studied Gilbert Strang's *"Linear Algebra and Its Applications"*, his explanation of Schur Form was not very clear. This led me to find out more about Schur decomposition to the point where I could implement it.

## 1 Theorem Statement

**Theorem 1** (Schur Decomposition)**.** *Let $A \in \mathbb{C}^{n\times n}$. There exists a unitary matrix $U \in \mathbb{C}^{n\times n}$ and an upper triangular matrix $T \in \mathbb{C}^{n\times n}$ such that:*

$$U^{-1}AU = T \quad \iff \quad A = UTU^{\dagger}$$

## 2 Proof and Method by Induction

### 2.1 Base Case

For $A \in \mathbb{C}^{1\times 1}$, $A = [a_{11}]$.

- Choose $U_1 = [1]$, its inverse is $U_1^{-1} = [1]$.
- Compute $U_1^{-1}AU_1 = [1][a_{11}][1] = [a_{11}] = T$.
- A $1 \times 1$ matrix is trivially upper triangular.

### 2.2 Inductive Hypothesis

If any $(n-1)\times(n-1)$ matrix $A_2$ can be decomposed using a unitary matrix $W_2$ such that:

$$W_2^{-1}A_2W_2 = T_{n-1}$$

where $T_{n-1}$ is an upper triangular matrix.

### 2.3 Inductive Step

Let $A$ be an arbitrary $n \times n$ matrix. We must reduce it to size $(n-1)\times(n-1)$ and link it to our hypothesis.

1. We use the Power Method (or general eigenvalue analysis) to find a dominant eigenvalue $\lambda_1$ and its corresponding non-zero eigenvector $\vec{u}_1$ such that $A\vec{u}_1 = \lambda_1\vec{u}_1$.
2. We use the Gram-Schmidt process to expand our eigenvector $\vec{u}_1$ to a set of orthonormal vectors $\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_n\}$.
3. Construct $U_1$: We place these vectors as the columns of $U_1$:

$$U_1 = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \ldots & \vec{u}_n \end{bmatrix}$$

We want to prove that $U_1^{-1}AU_1$ has zeros everywhere in the first column below $\lambda_1$.

$$AU_1 = \begin{bmatrix} A\vec{u}_1 & A\vec{u}_2 & \dots & A\vec{u}_n \end{bmatrix}$$

When a matrix multiplies another matrix, it acts on each column individually. For the first column, we get:

$$A\vec{u}_1 = \lambda_1 \vec{u}_1$$

For the other columns ($A\vec{u}_2$ through $A\vec{u}_n$), the outputs are vectors in space. Because $\{\vec{u}_1, \dots, \vec{u}_n\}$ forms a basis, any vector can be written as a linear combination of them:

$$A\vec{u}_j = x_{1j}\vec{u}_1 + x_{2j}\vec{u}_2 + \cdots + x_{nj}\vec{u}_n$$

Notice that:

$$U_1 \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \dots & \vec{u}_n \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = c_1\vec{u}_1 + c_2\vec{u}_2 + \cdots + c_n\vec{u}_n$$

This shows that multiplying $U_1$ by a column vector of constants is the same as making a linear combination of the columns of $U_1$.

- For the 1st column: $\lambda_1\vec{u}_1 + 0\vec{u}_2 + \cdots + 0\vec{u}_n$. The column vector of numbers we need to multiply $U_1$ by is $\begin{bmatrix} \lambda_1 & 0 & \dots & 0 \end{bmatrix}^T$.
- For the rest of the columns, $(A\vec{u}_j)$ corresponds to $\begin{bmatrix} x_{1j} & x_{2j} & \dots & x_{nj} \end{bmatrix}^T$.

Putting them side by side forms:

$$AU_1 = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \dots & \vec{u}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & x_{12} & \dots & x_{1n} \\ 0 & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & x_{n2} & \dots & x_{nn} \end{bmatrix}$$

So this is:

$$AU_1 = U_1 \begin{bmatrix} \lambda_1 & c_1^T \\ \mathbf{0} & A_2 \end{bmatrix}$$

Multiplying by $U_1^{-1}$:

$$U_1^{-1}AU_1 = \begin{bmatrix} \lambda_1 & c_1^T \\ \mathbf{0} & A_2 \end{bmatrix}$$

The inner submatrix $A_2$ has dimension $(n-1) \times (n-1)$. By our inductive hypothesis, we can recursively decompose $A_2$. There exists an $(n-1) \times (n-1)$ unitary matrix $W_2$ such that $W_2^{-1}A_2W_2 = T_{n-1}$ is upper triangular.

Define:

$$U_2 = \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & W_2 \end{bmatrix}, \quad U_2^{-1} = \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & W_2^{-1} \end{bmatrix}$$

We substitute into $U_2^{-1}(U_1^{-1}AU_1)U_2$:

$$\begin{aligned} U_2^{-1}(U_1^{-1}AU_1)U_2 &= \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & W_2^{-1} \end{bmatrix} \begin{bmatrix} \lambda_1 & c_1^T \\ \mathbf{0} & A_2 \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & W_2 \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 & c_1^T W_2 \\ \mathbf{0} & W_2^{-1}A_2W_2 \end{bmatrix} \end{aligned}$$

Since $W_2^{-1}A_2W_2 = T_{n-1}$:

$$U_2^{-1}(U_1^{-1}AU_1)U_2 = \begin{bmatrix} \lambda_1 & c_1^T W_2 \\ \mathbf{0} & T_{n-1} \end{bmatrix} = \text{upper triangular matrix}$$

The final upper triangular matrix $T$ is:

$$T = \begin{bmatrix} \lambda_1 & c_1^T W_2 \\ \mathbf{0} & T_{n-1} \end{bmatrix}$$

Since $T_{n-1}$ is upper triangular, this keeps the full matrix $T$ upper triangular.

The full unitary transformation matrix is:

$$U = U_1U_2U_3 \dots U_n$$

This is our final unitary matrix $U$.

Since we have proved that an $n \times n$ matrix can be decomposed if an $(n-1) \times (n-1)$ matrix can, and our base case works, we have proven Schur Decomposition by induction and derived a method to compute it.

# 3 Properties of the Unitary Product and Eigenvalues

To prove that the product of unitary matrices is unitary, note that for $U_1, U_2$:

$$(U_1U_2)^H = U_2^H U_1^H = U_2^{-1}U_1^{-1} = (U_1U_2)^{-1}$$

because $U_2^{-1}U_1^{-1}U_1U_2 = I$.

## 3.1 Eigenvalues on the Diagonal

The eigenvalues of $A$ are on $T$'s diagonal. This is because $A$'s eigenvalues and $T$'s eigenvalues are the same, and $T$'s eigenvalues lie on its diagonal.

Evaluating $\det(T - \lambda I)$:

$$\det(T - \lambda I) = \det \begin{bmatrix} \lambda_1 - \lambda & & * \\ & \lambda_2 - \lambda & \\ 0 & & \ddots \end{bmatrix} = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda)$$

Since the terms below $(\lambda_1 - \lambda)$ are 0, the determinant simplifies directly to the product of diagonal entries.

To prove that $A$'s eigenvalues are $T$'s eigenvalues:

$$\begin{aligned} \det(A - \lambda I) &= \det(UTU^{-1} - \lambda UU^{-1}) \\ &= \det(U)\det(T - \lambda I)\det(U^{-1}) \\ &= \det(UU^{-1})\det(T - \lambda I) \\ &= \det(T - \lambda I) \end{aligned}$$

Because $A$'s eigenvalues are equivalent to $T$'s eigenvalues, and $T$'s eigenvalues lie on its diagonal, $A$'s eigenvalues can be read directly off $T$'s diagonal.

## Further improvement of Schur

In the first step, we put one eigen vector and get the first column in T as lambda and all rest as 0. In fact, this has geometrically meanings that we want to explore with quotient space, as in the book LADR. Here, for this proof, we can actually calculate the whole eigen vectors for this lambda, say k of them. And we can put this k vectors as the first k in U, then we can get the first k columns in T having k by k matrix as $\lambda I$ and rest as 0 matrix. This prove will still work. Morevoer, for defective matrix, we can still find $\lambda$ still as eigen value of $A_1$. But this still would work.